



Catena-X
Automotive Network

IDTA 02035-6 : Digital Battery Passport - Part 6 Material Composition

Version 1.0.1

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SPECIFICATION

Submodel Template of the
Asset Administration Shell



Submodel Template

IDTA approved

- 100% AAS compliant
- Consistent & interoperable
- Released by the AAS experts

IDTA 02035-6

Imprint

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Version history

For a detailed change history see [Annex B. Changes to the submodel template](#).

Date	Version	Comment
May 2026	1.0.1	Bugfix Version, for a detailed change history see Annex.
February 2026	1.0	Release of the official Submodel Template published by IDTA – a joint result of the Model Expert Group comprising IDTA, Catena-X, and the Battery Pass Consortium.

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Chapter 1. General

1.1. About this document

This document is a part of an overall specification series [4]. Each part specifies the contents of a Submodel Template (SMT). The specifications of the Asset Administration Shell (AAS) are the basis for the Submodel Template specifications, see [3].

The target audience of the specification are developers and editors of technical documentation and manufacturer information, which are describing assets by means of the Asset Administration Shell (AAS) and therefore need to create a Submodel instance with a hierarchy of SubmodelElements. This document especially details on the question, which SubmodelElements with which semantic identification shall be used for this purpose.

This specification was created following the "semantic-driven workflow" as defined in [5] based on Aspect Models [6]. There is no central dictionary or repository for Aspect Models. In this specification the following sources are used for defining semantics:

- Aspect Models published at IDTA [7]: <https://github.com/admin-shell-io/smt-semantic-models>, models with namespace "io.admin-shell"
- Aspect Models published by the [BatteryPass Consortium](https://github.com/batterypass/BatteryPassDataModel) (closed project) [8]: <https://github.com/batterypass/BatteryPassDataModel>, models with namespace "io.BatteryPass"
- Aspect Models published at Tractus-X and used in standards published by [Catena-X](https://github.com/eclipse-tractusx/sldt-semantic-models) [9]: <https://github.com/eclipse-tractusx/sldt-semantic-models>, models with namespace "io.catenax"

1.2. Scope of the Submodel

This Submodel template aims to define the material composition data points of a Battery Passport conformant to DIN DKE SPEC 99100 and the corresponding EU regulations.

The battery passport consists of the following 7 parts:

Digital Battery Passport - Part 1: Digital Nameplate (IDTA-02035-1)
Digital Battery Passport - Part 2: Handover Documentation (IDTA-02035-2)
Digital Battery Passport - Part 3: Product Carbon Footprint (IDTA-02035-3)
Digital Battery Passport - Part 4: Technical Data (IDTA-02035-4)
Digital Battery Passport - Part 5: Product Condition (IDTA-02035-5)
Digital Battery Passport - Part 6: Material Composition (IDTA-02035-6)
Digital Battery Passport - Part 7: Circularity (IDTA-02035-7)

This specification is Part 5: Material Composition (IDTA_02035-6).

1.3. Relevant standards for the Submodel Template

This submodel template fulfills the requirements for dynamic data attributes as defined in DIN DKE SPEC 99100 [1]. DIN DKE SPEC 99100 "is based on the European Union and key Member States current regulatory requirements for battery passport information. Mandatory information for the battery passport as stated in the EU Battery Regulation (EU)2023/1542, Article77 and AnnexXIII, as well as the Ecodesign for Sustainable Products Regulation (ESPR), is supplemented by recommendations to increase sustainability and circularity. [1]"

This document is valid for all battery categories. Please be aware that for battery categories that have stronger requirements like industrial batteries with battery management systems etc. some of the data points are specified as optional although mandatory per regulation.

1.4. Explanations on used UML diagrams

For clarity and an improved legibility readers suggested to go through this section at first before reading the following chapters.

UML diagrams feature box-like elements, called "classes". These classes, typically Submodels, SubmodelElementCollections or SubmodelElementLists, typically feature a set of Properties or further SubmodelElements. These elements can have specific cardinalities.

The single classes are hierarchally organized by aggregation relations, these can be seen as "contains" relation.

For a further overview on UML diagrams please refer to [2] and [3].

Further details about used table formats please refer to Annex A Explanations.

Chapter 2. Information set for Submodel “MaterialComposition”

2.1. General

The "Material Composition" Submodel Template is part of the specification series for the Battery Passport.

Property specification

See [clause 3 "Information structures and attributes"](#).

2.2. Overview UML model

The SubmodelElements described in section 3 are structured in the following way (see [Figure 1](#)):

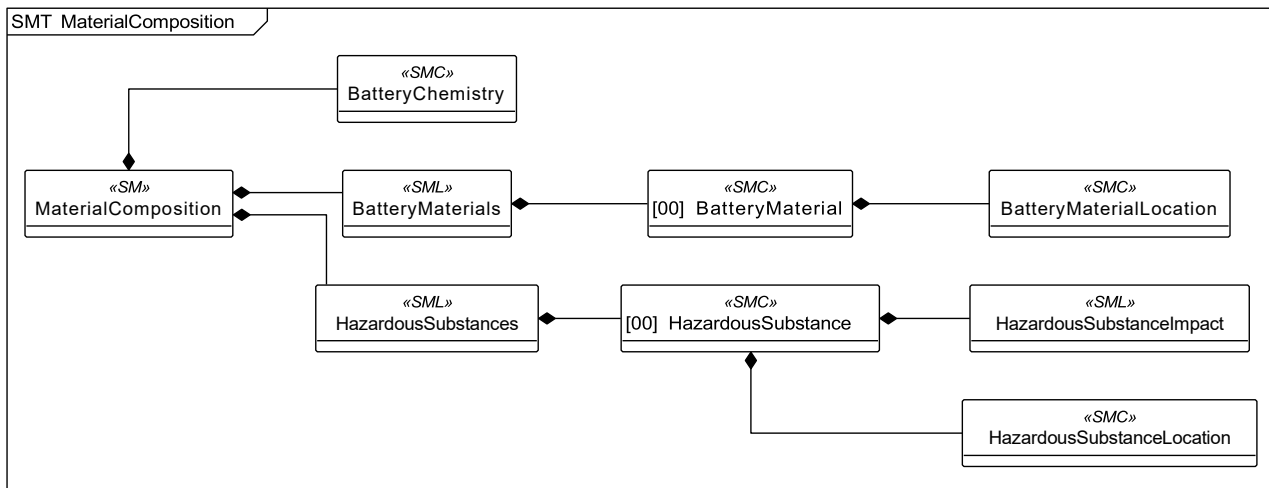


Figure 1. UML overview

Chapter 3. Information structures and attributes

3.1. Properties of the Submodel “Material Composition”

Figure 2 shows the UML-diagram defining the relevant properties which need to be set.

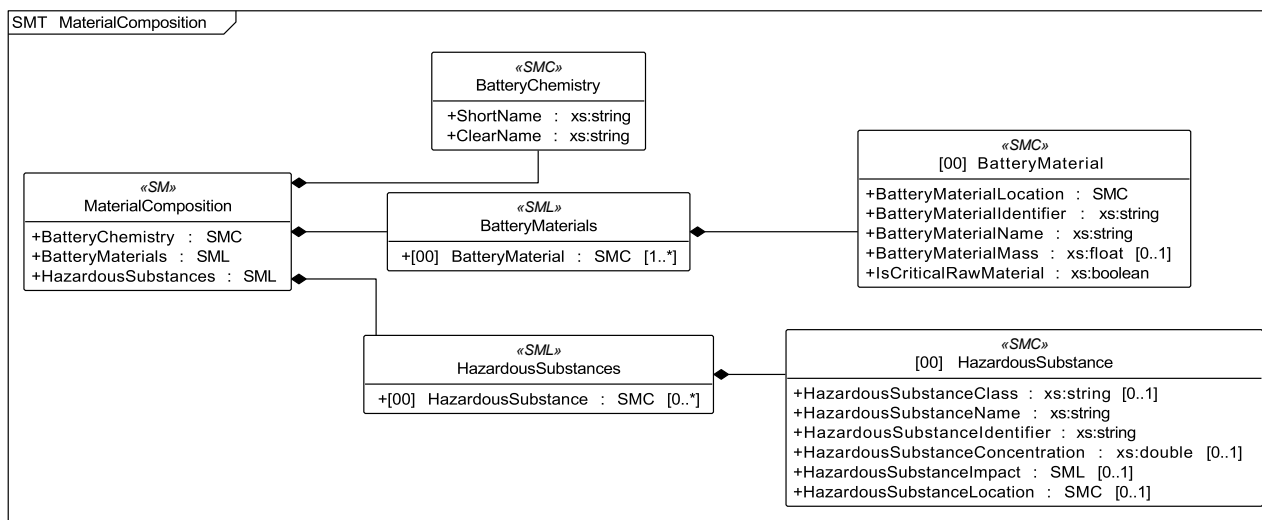


Figure 2. UML-Diagram for Submodel "MaterialComposition"

idShort:	MaterialComposition		
Class:	Submodel		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#MaterialComposition		
Parent:	defaultAdminShell		
Explanation:	Mandatory data: Battery chemistry; critical raw materials; materials used in the cathode, anode, and electrolyte; hazardous substances; impact of substances on the environment and on human health or safety.		
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[SMC]	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#batteryChemistry	2 elements	1
BatteryChemistry	Composition of a product in general terms by specifying the cathode and anode active material as well as electrolyte. DIN KE Spec 99100 chapter reference: 6.5.2		

[SML] BatteryMaterials	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#batteryMaterials "Component materials used" (No. 17.a-c): Naming the materials (as a composition of substances) in cathode, anode, electrolyte according to public standards, including specification of the corresponding component (i.e., cathode, anode, or electrolyte). We suggest a reporting threshold of 0.1 % weight by weight. DIN DKE Spec 99100 chapter reference: 6.5.3-6.5.4	1 1 elements	1
[SML] HazardousSubstances	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#hazardousSubstances "Hazardous substances" (No 20.a-e): Name (agreed substance nomenclature, e.g. IUPAC or chemical name) all hazardous substance (as 'any substance that poses a threat to human health and the environment'). Suggested above 0.1 % weight by weight within each (sub-)component. DIN DKE Spec 99100 chapter reference: 6.5.4 - 6.5.6	1 1 elements	1

3.2. Properties of the SMC "BatteryChemistry"

Figure 3 shows the UML-diagram for [SMC BatteryChemistry](#).

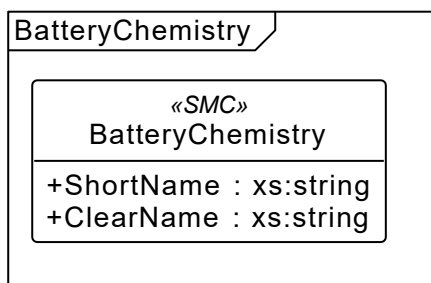


Figure 3. UML-Diagram for SMC "BatteryChemistry"

Data points related to battery chemistry.

idShort:	BatteryChemistry
Class:	SubmodelElementCollection
semanticId:	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#batteryChemistry
Parent:	MaterialComposition
Explanation:	
Element details:	-

[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop]	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#shortName	[String]	1
ShortName	Short name.	NMC	
[Prop]	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#clearName	[String]	1
ClearName	Clear name.	Lithium nickel manganese cobalt oxides	

3.3. Properties of the SML "BatteryMaterials"

Figure 4 shows the UML-diagram for [SML batteryMaterials](#).

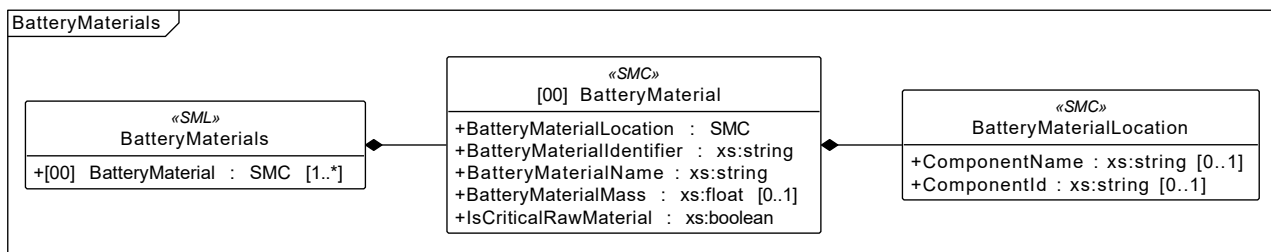


Figure 4. UML-Diagram for SML "BatteryMaterials"

Data points related to a list of battery materials.

idShort:	BatteryMaterials		
Class:	SubmodelElementList		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#batteryMaterials		
Parent:	MaterialComposition		
Explanation:			
Element details:	orderRelevant=No, typeValueListElement=SubmodelElementCollection		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[SMC]	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#BatteryMaterial	[]	1..*
BatteryMaterial	Material of battery.	5 elements	

3.4. Properties of the SML "HazardousSubstances"

Figure 5 shows the UML-diagram for [SML hazardousSubstances](#).

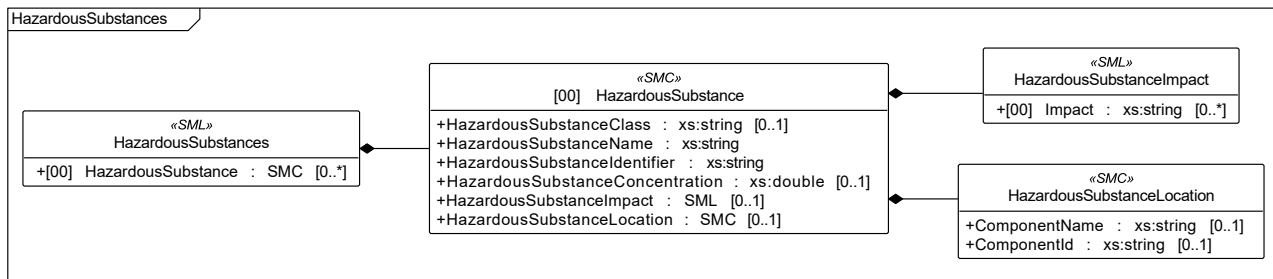


Figure 5. UML-Diagram for SML "HazardousSubstances"

Data points related to a list of hazardous substances.

idShort:	HazardousSubstances		
Class:	SubmodelElementList		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#hazardousSubstances		
Parent:	MaterialComposition		
Explanation:			
Element details:	orderRelevant=No, typeValueListElement=SubmodelElementCollection		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[SMC]	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#HazardousSubstance	␣ 6 elements	0..*
HazardousSubstance	"Hazardous substance" (No 20.a-e): Name (agreed substance nomenclature, e.g. IUPAC or chemical name) a hazardous substance (as 'any substance that poses a threat to human health and the environment'). Suggested above 0.1 % weight by weight within each (sub-)component. DIN DKE Spec 99100 chapter reference: 6.5.4 - 6.5.6		

3.5. Properties of the SMC "BatteryMaterial"

Figure 6 shows the UML-diagram for [SMC BatteryMaterial](#).

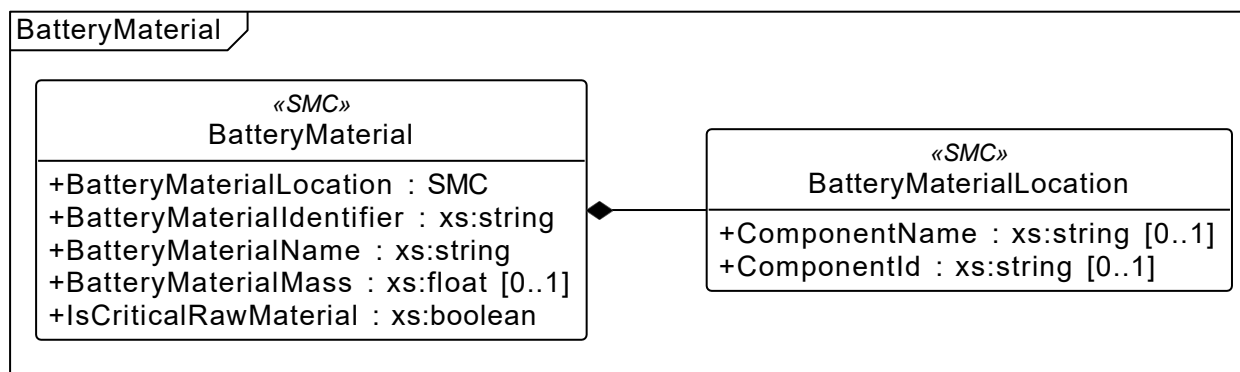


Figure 6. UML-Diagram for SMC "BatteryMaterial"

Data points related to a battery material.

idShort:	BatteryMaterial		
Class:	SubmodelElementCollection		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#BatteryMaterial		
Parent:	BatteryMaterials		
Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[SMC] BatteryMaterial Location	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#batteryMaterialLocation Battery component that relates to the material.	[] 2 elements	1
[Prop] BatteryMaterial Identifier	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#batteryMaterialIdentifier supplementalSemanticId: https://www.cas.org/cas-data/cas-registry , urn:irdi:0173-1#02-ABH034#003 Registration number of an international designation standard (CAS = Chemical Abstracts Service) for chemical substances. For each chemical substance registered in the CAS database (also biosequences, alloys, polymers) there is a unique CAS number.	[String] 7439-93-2	1

[Prop] BatteryMaterial Name	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#battery MaterialName supplementalSemanticId: urn:irdi:0173-1#02-AAN920#003 Clear name of Material. Alphanumeric character string which describes the object.	[String] Lithium	1
[Prop] BatteryMaterial Mass	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#battery MaterialMass Weight of component material in Kilogram.	[Float] 52	0..1
[Prop] IsCriticalRawM aterial	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#isCritical RawMaterial The battery passport must contain information on the critical raw materials present in the battery. The information on the critical raw materials must also be provided on the battery label. Per Annex VI, Part A(10), critical raw materials must be reported if present in the battery in a concentration of more than 0,1 % weight by weight.	[Boolean] true	1

3.6. Properties of the SMC "HazardousSubstance"

Figure 7 shows the UML-diagram for [SMC HazardousSubstance](#).

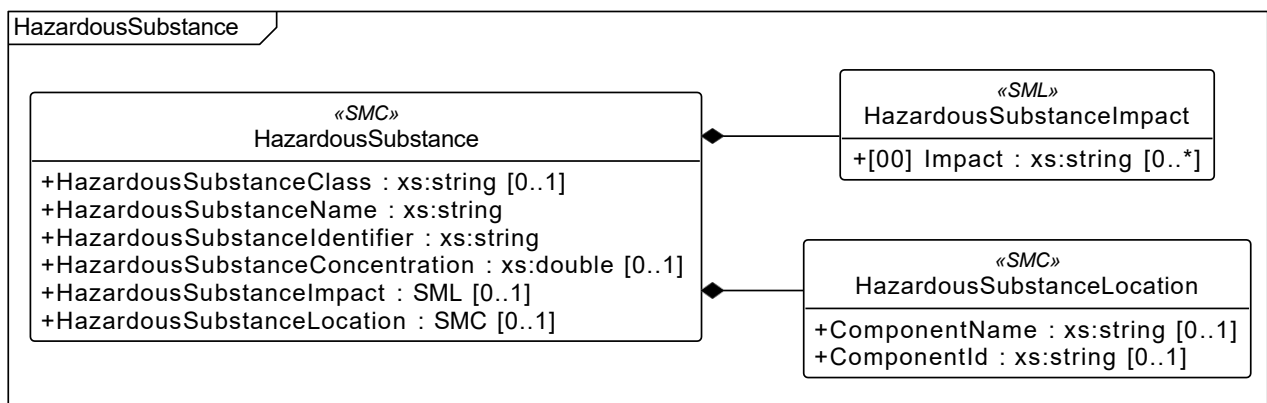


Figure 7. UML-Diagram for SMC "HazardousSubstance"

Data points related to a hazardous substance.

idShort:	HazardousSubstance
Class:	SubmodelElementCollection
semanticId:	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#HazardousSubstance

Parent:	HazardousSubstances		
Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop] HazardousSubstanceClass	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#hazardousSubstanceClass Battery Regulation narrows reporting to substances falling under defined hazard classes and categories of the CLP regulation.	[String] SkinCorrosionOrIrritation	0..1
[Prop] HazardousSubstanceName	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#hazardousSubstanceName Clear name of hazardous substance.	[String] Cobalt (Co)	1
[Prop] HazardousSubstanceIdentifier	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#hazardousSubstanceIdentifier supplementalSemanticId: https://www.cas.org/cas-data/cas-registry CAS identifier of hazardous substance.	[String] 346417-97-8	1
[Prop] HazardousSubstanceConcentration	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#hazardousSubstanceConcentration Concentration of hazardous substance.	[Double] 20	0..1
[SML] HazardousSubstanceImpact	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#hazardousSubstanceImpact Impact statements based on, e.g., REACH or GHS for all hazard classes applicable to substances in the battery.	[] 1 elements	0..1
[SMC] HazardousSubstanceLocation	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#hazardousSubstanceLocation Hazardous substances (No. 19-23): Location on a (sub-)component-level of all hazardous substances (as 'any substance that poses a threat to human health and the environment'). Suggested via a unique identifier or nomenclature.	[] 2 elements	0..1

For hazardousSubstanceClass the following values are allowed:

- "AcuteToxicity"

- "SkinCorrosionOrIrritation"
- "EyeDamageOrIrritation"

3.7. Properties of the SMC "BatteryMaterialLocation"

Figure 8 shows the UML-diagram for [SMC batteryMaterialLocation](#).

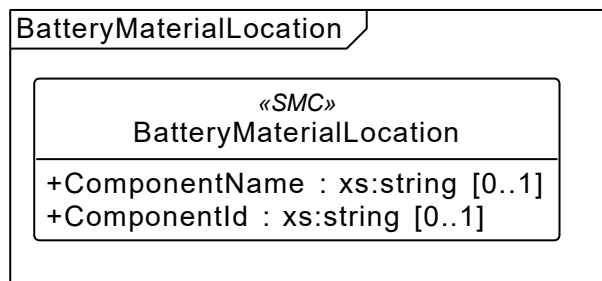


Figure 8. UML-Diagram for SMC "BatteryMaterialLocation"

Data points related to battery material location.

idShort:	BatteryMaterialLocation		
Class:	SubmodelElementCollection		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#batteryMaterialLocation		
Parent:	BatteryMaterial		
Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop]	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#componentName	[String]	0..1
ComponentName	Name of component.	Anode	
[Prop]	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#componentId	[String]	0..1
ComponentId	Identifier of the component.		

3.8. Properties of the SML "HazardousSubstanceImpact"

Figure 9 shows the UML-diagram for [SML HazardousSubstanceImpact](#).

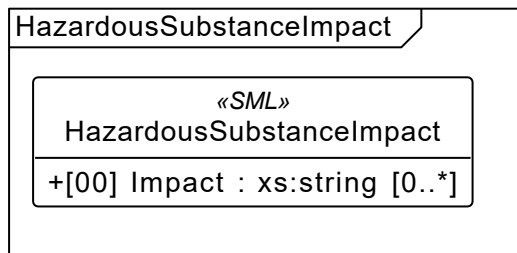


Figure 9. UML-Diagram for SMC "HazardousSubstanceImpact"

Data points related to a list of hazardous substances impacts.

idShort:	HazardousSubstanceImpact		
Class:	SubmodelElementList		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#hazardousSubstanceImpact		
Parent:	HazardousSubstance		
Explanation:			
Element details:	orderRelevant=No, typeValueListElement=Property, valueTypeListElement=xs:string		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop]	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#Impact	[String]	0..*
Impact	Impact.	May cause an allergic skin reaction	

To ensure clear and consistent communication of risks, the impact is reported through Hazard (H) phrases and Precautionary (P) statements stipulated in the CLP Regulation (EC No 1272/2008).

Physical hazards include:

- Highly flammable liquid and vapour
- May be corrosive to metals

Health hazards include:

- May cause an allergic skin reaction
- May cause allergy or asthma symptoms or breathing difficulties if inhaled
- Suspected of causing cancer
- May damage fertility

Environmental hazards include:

- Toxic to aquatic life with long lasting effects

3.9. Properties of the SMC

"HazardousSubstanceLocation"

Figure 10 shows the UML-diagram for SMC HazardousSubstanceLocation.

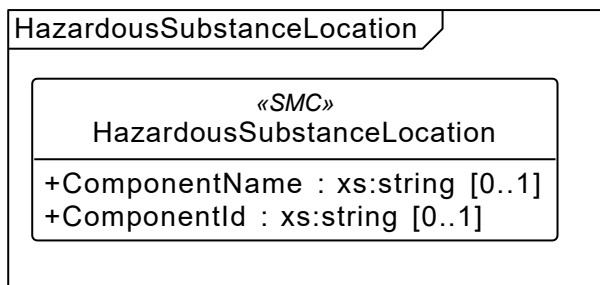


Figure 10. UML-Diagram for SMC "HazardousSubstanceLocation"

Data points related to hazardous substance location.

idShort:	HazardousSubstanceLocation		
Class:	SubmodelElementCollection		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#hazardousSubstanceLocation		
Parent:	HazardousSubstance		
Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop]	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#componentName	[String]	0..1
ComponentName	Name of component.	Anode	
[Prop]	urn:samm:io.admin-shell.idta.batterypass.material_composition:1.0.1#componentId	[String]	0..1
ComponentId	Identifier of the component.	1234-344	

Annex A. Explanations on used table formats

1. General

The used tables in this document try to outline information as concise as possible. They do not convey all information on Submodels and SubmodelElements. For this purpose, the definitive definitions are given by a separate file in form of an AASX file of the Submodel template and its elements.

2. Tables on Submodels and SubmodelElements

For clarity and brevity, a set of rules is used for the tables for describing Submodels and SubmodelElements.

- The table heads abbreviate 'cardinality' with 'card'.
- The tables often place two informations in different rows of the same table cell. In this case, the first information is marked out by sharp brackets [] from the second information.
- The types of SubmodelElements are abbreviated (see [Table 1](#)):

Table 1. Abbreviations for SubmodelElements

SME type	SubmodelElement type
Blob	Blob
Cap	Capability
Ent	Entity
Evt	Event
File	File
MLP	MultiLanguageProperty
Opr	Operation
Prop	Property
Range	Range
Ref	ReferenceElement
Rel	RelationshipElement
RelA	AnnotatedRelationshipElement
SMC	SubmodelElementCollection
SME	SubmodelElement
SML	SubmodelElementList

- Multi-language strings are represented by the text value, followed by '@'-character and the ISO 639 language code: example@EN.
- The [valueType] is only given for Properties.

Annex B. Changes to the submodel template

General

This annex lists the changes from version to version of the Submodel, together with major changes in the overall document.

Changes Version 1.0.1

Major Changes

- CHANGE - make BatteryMaterialMass optional (#83)
- CHANGE - change measurement unit of BatteryMaterialMass from gram to kilogram (#85)
- CHANGE - make HazardousSubstanceClass, HazardousSubstanceClass, HazardousSubstanceConcentration, HazardousSubstanceImpact and HazardousSubstanceLocation optional in HazardousSubstance (#83)
- ADDED - add qualifier "EitherOr" to ComponentId and ComponentName in BatteryMaterialLocation and HazardousSubstanceLocation (as explained in definition of the concepts)

Minor Changes

- add examples in text for hazardous substance impact
- add unitId for Measurements, i.e. BatteryMaterialMass and HazardousSubstanceConcentration
- add additional example values
- editorial changes in descriptions
- editorial changes

Changes Version 1.0

- First Version conformant to DIN DKE SPEC 99100

Bibliography

- [1] DIN DKE SPEC 99100, "Requirements for data attributes of the battery passport". February 2025.
- [2] "OMG Unified Modeling Language (OMG UML)", Formal/2017-12-05, Version 2.5.1. December 2018. [Online] Available: <https://www.omg.org/spec/UML/>
- [3] "Specification of the Asset Administration Shell", Publisher: Industrial Digital Twin Association (IDTA). [Online]. Available: <https://industrialdigitaltwin.org/en/content-hub/aasspecifications>
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- [6] "Semantic Aspect Meta Model (SAMM)", V2.2.0. [Online]. Available: <https://eclipse-esmf.github.io/samm-specification/2.2.0/index.html>
- [7] "Semantic Aspect Models - smt-semantic-models", Publisher: Industrial Digital Twin Association (IDTA). [Online]. Available: <https://github.com/admin-shell-io/smt-semantic-models>
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- [9] "Semantic Aspect Models - Tractus-X - sltd-semantic-models", [Online]. Publisher: Catena-X. [Online]. Available: <https://github.com/eclipse-tractusx/sltd-semantic-models>